

Overview

Jitsu is a last-mile delivery company and technology platform. Our clients are e-commerce retailers who seek alternatives to the USPS, traditional national carriers like UPS and FedEx, and incumbent regional carriers such as OnTrac.

Jitsu is asset-light. We do not own delivery vehicles. Our facilities are leased, and our smaller deployment-only locations ("mobile hubs") are rented month-to-month. This lowers fixed operating costs and allows us to expand or contract our footprint fluidly.

Jitsu achieved SOC 2 Type II compliance in April 2024 (audited by Sensiba LLP), establishing an enterprise-ready security posture for shipper and partner requirements.

Key Terminology

This section defines terms used throughout this document.

Shipment - A single parcel that Jitsu receives from a client and delivers to a recipient (a.k.a. customer). Each shipment has a geocoded delivery address.

Route - An ordered sequence of stops assigned to a single driver. A route consists of one or more pickup stops (where the driver loads parcels at a facility) followed by delivery stops (where the driver delivers parcels to recipients). Also called an "assignment"; these terms are interchangeable, but we use "route" throughout this document.

Zone - A geographically contiguous neighborhood with high delivery density. Zones are used both for routing (routes are created within zones) and for driver booking (drivers can reserve delivery opportunities in specific zones).

Stem miles - The distance a driver must travel from the deployment facility to their first delivery stop, and from their last delivery stop back. Minimizing stem miles is a key optimization objective.

OTD (on-time delivery) - The percentage of shipments delivered within their promised delivery window. OTD is one of Jitsu's primary service quality metrics.

Courier - As Jitsu uses the term, refers specifically to a Delivery Service Provider (DSP) or a third-party gig platform driver (3P). It never refers to an Independent Contractor (IC) driver. This is a Jitsu-specific usage.

SLA (service level agreement) - Defines the delivery speed commitment for a shipment. Jitsu typically offers same-day, next-day, and standard (deferred) service levels.

Tour cost - The base pay for a route before any bonuses.

Markets and Facilities

Jitsu operates in approximately 29 US regions/markets: Indianapolis (IND), Houston (HOU), Dallas (DFW), Austin (AUS), St. Louis (STL), Chicago (CHI), Milwaukee (MKE), Detroit (DTW), SF Bay Area (SFO), Los Angeles (LAX), Sacramento (SMF), San Diego (SDLAX), Las Vegas (LAS), Seattle (SEA), Stamford (HPN), New Haven (HVN), Cincinnati (CVG), New York (JFK), Baltimore (BWI), New Jersey (EWR), Philadelphia (PHL), Pittsburgh (PIT), Washington DC (IAD), Richmond (RIC), Phoenix (PHX), Norfolk (ORF), Columbus (CMH), Portland (PDX), Cleveland (CLE).

Each region is served by one or more facilities. A single facility can serve one or more of the following functions:

- **Sortation center:** receives shipments in a process called "inbound," scans and labels them, and physically sorts them into their designated route positions (for locally-bound shipments) or into containers bound for other markets (for cross-dock shipments).
- **Delivery station:** a location where pre-sorted routes are staged and deployed to drivers in a process called "outbound."
- **Cross-dock center:** receives shipments bound for multiple distant markets and sorts them into containers for middle-mile transportation to those markets.

A **mobile hub** is a lightweight, month-to-month rented delivery station located in a satellite neighborhood away from the main sortation center. Mobile hubs reduce stem miles by placing deployment points closer to clusters of deliveries. The decision to open a mobile hub on a given day can be made dynamically - as late as the day before - based on predicted geographic density of shipments.

Workforce

Driver Fleet

Jitsu does not employ drivers directly. We use a mixed fleet composed of several provider types:

- **Independent Contractors (ICs):** individual drivers who book delivery opportunities through the Jitsu driver app. ICs generally deliver within specific zones and select routes through a booking process described later in this document.
- **Delivery Service Providers (DSPs):** contracted delivery companies that handle routes in negotiated geographic areas at rates based on parcel count, parcel type, and route mileage. DSPs are not bound by zone boundaries.

- **Third-party gig platforms (3P):** labor platforms such as WorkWhile, InstaWork, and Wonolo. 3P drivers are used in exigent situations when IC and DSP coverage is insufficient - for example, due to driver supply shortages, unexpected volume surges, or late-deploying routes caused by sortation delays. Like DSPs, 3P drivers are not bound by zone boundaries.

DSPs and 3P drivers are collectively referred to as "couriers." ICs are never called couriers.

The mix of provider types varies by region and depends on local driver supply. Jitsu typically launches a new market with DSPs to guarantee route coverage, then transitions to ICs as driver supply grows.

Facility Staff

Most workers in Jitsu's facilities are contingent workers hired on a temporary basis from labor platforms. This is consistent with the asset-light model and allows staffing to scale with volume.

Client Lifecycle

Sales

Clients are prospected by the Sales organization. A sales development representative (SDR) scouts potential accounts and engages Account Executives once a prospect is interested.

Pricing and Network Design

The pricing team, in conjunction with the linehaul and sales teams, constructs a network design for each client. The network design details the most cost-efficient way for the client to get their products to Jitsu. This first shipment leg is called the "first mile" and is sometimes orchestrated by Jitsu and sometimes by the client or their freight partners. Time- and cost-efficient first mile is often one of the biggest challenges clients face when deciding to work with Jitsu, because unlike national carriers we do not have induction points and a fully connected network nationwide.

Implementation

The Implementation team determines the best way for a client to connect their systems to Jitsu's and then performs and tests the integration. Jitsu offers a broad REST API documented at <https://docs.gojitsu.com/>, including near real-time webhooks. We also support inbound SFTP for shipment feeds and outbound SFTP for tracking updates and invoicing. We integrate with post-transaction visibility partners like AfterShip and Narvar, as well as spend management and auditing platforms like Intelligent Audit, StrategIQ Commerce, Platinum Circle, Shiplab, and similar.

The Implementation team also takes client contract pricing and codes it into rate tables for use with clients' rate-shopping systems via our real-time rating API. Rate tables are powered by Jitsu's pricing DSL (domain-specific language) - a purpose-built formula language that non-engineers can read and modify. Pricing formulas resemble spreadsheet expressions (for example: `baseRate + (boxes * perBoxRate) + $IF(isRemote, remoteSurcharge, 0)`) and support arithmetic, conditionals, pattern matching, tiered pricing, volume discounts, and zone-based rate structures. Sales can configure client-specific pricing without engineering involvement, and Finance can audit formula logic directly. Each price calculation produces a full trace of inputs, intermediate values, and final price for audit and dispute resolution.

Client Success

The Client Success team manages accounts after clients go live. They handle day-to-day escalations, new regional rollouts, operational changes, and general account health. They are the clients' advocates by proxy and raise concerns with Operations or Engineering when issues arise.

Operations Organization

The Operations team handles linehaul, facilities, transportation, call center, and dispatch. Each function is handled by a dedicated team.

Linehaul

The Linehaul team manages Jitsu-orchestrated first-mile freight lanes as well as middle-mile transfers between Jitsu facilities. Middle mile is the transportation of shipments from one Jitsu facility to another - for example, from a sortation center to a mobile hub, or from a cross-dock center to a sortation center in another region.

Facilities

The Facilities team operates sortation centers, delivery stations, and cross-dock centers. They are responsible for all on-site inbound and outbound operations and supervision of temporary workers.

Transportation

The Transportation team includes driver services, planners, coordinators, and dispatchers. Their collective responsibility is ensuring Jitsu has enough drivers, that drivers are compliant with law and Jitsu's rules, that routes are cost- and time-efficient, and that drivers are supported on delivery day.

- **Driver Services** handles IC driver recruitment, background checks (automated via Turn.io with FCRA-compliant consent management), planning, and discipline. Discipline uses a graduated suspension system with five levels: complete blackout (blocks all route access for serious violations like fraud or safety issues), delayed access (driver can only book after a time delay, missing the best routes), reduced capacity (limits concurrent bookings while maintaining some earning opportunity), and lesser penalties for minor infractions. Each suspension type has a clear driver-facing explanation so drivers know why they are restricted and how to resolve it. Driver Services also handles general communications to drivers about significant operational or software changes.
- **Transportation Planners** have two major roles: (1) finding and negotiating with DSPs to cover more difficult delivery areas at cost-efficient rates, and (2) using Jitsu's proprietary routing app to plan optimal routes for a given market on a given day, considering expected volume and expected vehicle fleet composition.
- **Transportation Coordinators** work proactively with drivers to encourage route uptake and handle issues before they escalate.
- **Dispatchers** handle driver-reported issues reactively and periodically scan routes for drivers at risk of not delivering on time. They use the Jitsu Dispatch app.

Shipment Lifecycle

The remainder of this document follows a shipment through the Jitsu system, from first-mile arrival through delivery.

First Mile

Shipments arrive at Jitsu facilities via first-mile transportation - either orchestrated by Jitsu's Linehaul team or arranged by the client or their freight partners.

Inbound and Sortation

Inbound begins when first-mile or middle-mile trucks arrive at a sortation center. Shipments are unloaded and placed on conveyor belts, where they are scanned and labeled with a sort label. The sort label indicates either:

- A **route position**, for locally-bound shipments (called "final sort"). These shipments will be delivered from this facility's market.
- A **region code**, for shipments being cross-docked (called "regional sort"). These shipments are bound for a different Jitsu market.

Final-sorted shipments are placed into their designated locations on the sort floor: large parcels go onto pallets arranged in a grid layout, with each pallet position corresponding to a route;

small parcels go into bins. Cross-docked (regionally sorted) shipments go into a designated container or loading area for their destination market.

By the time physical sorting begins, the routing engine has already produced routes for the day (see "Routing" below). The sort labels direct shipments to the correct physical positions for those pre-computed routes.

For shipments that arrive unexpectedly and were not included in the original routing solution, Jitsu can "sprinkle" them - inserting them into an optimal position on an existing route that has capacity. To the sortation worker, a sprinkled shipment looks like any other pre-routed shipment; this keeps the sort line moving without delays. Sprinkling is described in more detail under "Routing."

After sorting is complete (a state called "locked"), the Facilities team prepares cross-docked shipments for middle-mile transportation and wraps final-sorted pallets so they can be safely moved by pallet jack to the outbound staging area.

Cross-Docking

Cross-dock centers and sortation centers with cross-dock capability receive shipments bound for multiple distant markets. During inbound, these shipments are labeled with a region code and sorted into containers designated for each destination market. After sort, the Linehaul team arranges middle-mile transportation to move these containers to the appropriate destination facilities.

Sortation Layout

The sortation layout defines the physical arrangement of pallet positions and bins on the sort floor. The layout can be generated automatically by Jitsu's constraint-based layout optimizer or created manually by a sortation lead (sort lead).

The automatic layout optimizer applies several strategies:

- Routes with the farthest delivery distances from the facility, or routes that must deploy earliest (such as those bound for mobile hubs), are placed closest to the dock doors on the outer edge of the layout grid.
- Routes are grouped geographically so that drivers who want to take multiple routes can select from nearby pallet positions.
- Routes too large to fit in a single pallet position are automatically split across multiple positions.
- Routes destined for mobile hub delivery via middle-mile truck are stacked for optimal space utilization in the transport vehicle (14-foot van, 26-foot truck, or 53-foot truck).
- New routes created mid-sort (for example, due to unexpected volume) are automatically added to the layout and to the advance booking session.

Multi-Cycle Operations (MCO)

To maximize facility utilization, Jitsu's sortation system supports multi-cycle operations. In MCO, a second sort cycle can begin while the first cycle's routes are still being outbound. This works within fixed space by allowing sorted pallets and bins from the first cycle to be "extracted" and staged for outbound, freeing those positions for the second cycle's routes.

Routing

Routing is the process of creating a set of routes for a given market on a given day. It involves predicting what shipments will need delivery, determining where they will go, and computing cost-optimized routes for a heterogeneous driver fleet.

Predicting Inbound Volume

Jitsu relies on a combination of client manifests and predictive models to determine which shipments to include in a day's routing problem.

Some clients - particularly subscription and meal-kit companies with rigid manufacturing and first-mile schedules - provide accurate manifests. These clients form the "backbone" of routing solutions but are the minority. Most e-commerce clients provide manifests that are at best loosely suggestive of actual inbound volume.

Because of manifest inaccuracy, Jitsu maintains predictive models that use historic trends, shipment first-mile origin, typical lag time from label generation to physical arrival, and other client-specific patterns to forecast inbound quantity.

Quantity alone is not sufficient; routing also requires geographic distribution of deliveries. Jitsu uses historical delivery data to estimate the expected geographic spread of shipments from clients with unreliable manifests.

To account for unpredictable volume, Jitsu creates "guessed" shipments - synthetic pseudo-shipments placed at the locations and in the quantities our models predict are most likely to need delivery. Guessed shipments function like real shipments for route creation. When a real shipment for that client is physically sorted, the system replaces the nearest guessed shipment with the real one and discards the guess. This keeps route shipment counts stable and as close to optimal as possible.

Service Levels and Deferred Inventory

Along with predicted volume, the routing engine considers each shipment's service level and contractual SLA. Day-definite deliveries (same-day, next-day) form the backbone of routing problems because they are hard-constrained to a specific delivery date. Deferred (standard) shipments offer flexibility: they can be slotted in where they improve route density or best match expected driver supply.

Transportation planners can view previously received shipment "inventory" in the Inbound app and use this information to construct routing problems best suited for the day's needs.

Route Creation

The routing engine first stratifies shipments by optimal vehicle-size allocation:

- **Super routes:** extra-large routes for the largest vehicle drivers.
- **Minivan/large routes:** for drivers with SUVs, wagons, or similar large vehicles.
- **DSP routes:** deliveries contractually allocated to meet weekly DSP volume minimums.
- **Zone routes:** standard IC routes within a geographic delivery zone.
- **LITE routes:** smaller routes for smaller vehicles within the immediate catchment of the delivery station.

This allocation takes advantage of natural delivery density to minimize total vehicle count. Combined with the mobile hub strategy, it also minimizes stem miles.

The routing engine then semi-automatically clusters shipments using a proprietary clustering strategy and generates an initial solution by inserting shipments into routes in an order that minimizes the cost of not being able to place each shipment in its best possible position (a technique that prioritizes difficult-to-place shipments first). It then iteratively improves the solution using "ruin and recreate" optimization - selectively dismantling portions of routes and rebuilding them - with proprietary cost functions and constraints, ultimately arriving at a cost-optimized routing solution that respects time and volume constraints of the anticipated fleet.

After a transportation planner selects a routing solution, the advance booking session is updated with the actual routes, and any additional booking tickets needed beyond the forecasted number are issued. At this time, drivers who were previously withheld from booking due to low driver rating scores are enabled for the session.

Sprinkling and Redelivery

For unexpected shipments sorted after routing is complete, Jitsu can "sprinkle" them by inserting each shipment into an optimal position on an existing route with remaining capacity. To the sortation worker, sprinkled shipments appear identical to pre-routed shipments, which keeps sort throughput high.

Shipments that cannot be sprinkled (no suitable route has capacity) are called "no routes." For these, Jitsu's redelivery app uses a regret-based optimization algorithm to find the best placement across all active routes globally.

Mobile Hub Routing

Based on the geographic distribution of shipments, the routing engine can designate clusters of routes around mobile hubs. In parallel, the Linehaul team arranges middle-mile trucks sized to the anticipated volume. This decision can be made dynamically, as late as the day before delivery, to account for last-minute volume fluctuations.

Driver Booking

Jitsu uses several booking mechanisms to match drivers with routes, escalating from advance booking to direct booking to third-party coverage as needed.

Advance Booking

Jitsu offers advance booking to IC drivers in good standing based on their driver ratings. Driver ratings are calculated from recent performance: on-time pickup, on-time delivery, route completion percentage, and (planned) delivery-within-window compliance reflecting stall time minimization.

The system uses statistics from prior weeks' IC routes to forecast how many delivery opportunities (tickets) to offer in advance - up to a week before delivery day. Tickets are made available in specific delivery zones. The system offers a percentage of anticipated IC opportunities in advance, holding back the remainder for day-of booking once actual route counts and DSP allocations are known.

The advance booking session is updated throughout the week as ticket forecasts change and actual routes materialize from the routing engine.

Drivers who abuse advance booking by canceling late are penalized: they cannot book routes as early as drivers with higher compliance scores.

Direct Booking

When advance booking does not yield full coverage, and before escalating to 3P drivers, Jitsu advertises routes directly (called "direct booking"). In contrast to ticket booking - where drivers see only the zone, pay range, and shipment count range - direct booking shows exact pricing and route location details, reducing uncertainty for prospective drivers. When coverage is still difficult, Jitsu may add a bonus on top of the base route pay (tour cost).

Third-Party Escalation

If IC and DSP coverage remains insufficient, 3P drivers from gig platforms are engaged as a last resort.

Pickup Slots and Yard Management

Each delivery station has a defined set of pickup slots - approximately 30-minute windows that drivers reserve alongside their booked ticket or route. Pickup slots correspond one-to-one with available tickets and routes and cover IC, DSP, and 3P drivers.

Delivery stations also have a parking slot map. Drivers choose or are assigned a parking spot upon arrival. Drivers arriving early for their pickup slot may be placed on a waitlist for parking. This enables yard management. Drivers who consistently arrive outside their expected pickup slot receive lower ratings and are offered delivery opportunities later.

Outbound

Outbound begins when drivers who have booked delivery opportunities arrive at a delivery station.

IC drivers check in by presenting their ID. After check-in, they are assigned a parking spot and select a route through "hybrid booking" - a system that gives drivers a choice among a constrained set of staged routes selected by the Outbound team. This is a hybrid of "driver's choice" (drivers pick) and "Jitsu's choice" (the available options are curated). Selecting a route with a ticket binds the ticket to that specific route.

DSP and 3P drivers arrive with routes already assigned to them.

These processes are collectively designed to maximize flow from driver arrival to departure. Jitsu tracks driver wait time as a key KPI because unpaid waiting time is one of the biggest sources of driver frustration.

Once a driver has a route, the outbound clerk brings the driver their pallet (large parcels) and any bin bags (small parcels). The driver scans and loads each parcel, reviews the route, accepts it, and departs. From this point, the driver's progress is monitored by Dispatch.

Jitsu Apps Used in Outbound

- **Jitsu Drive** ("the driver app"): the proprietary mobile app used by drivers for all booking, pickup, delivery, return, and payment operations.
- **Jitsu Outbound** ("the outbound app"): used by the Outbound team in conjunction with the Outbound Dashboard web app to track expected drivers, check them in, stage routes, assist with loading, and handle exceptions such as missorted packages or drivers leaving without checking out.

Delivery

Geofencing

Certain events at Jitsu are geofenced. Driver warehouse arrival (check-in) is geofenced so drivers can only get a parking slot and claim their route when physically near the arrival yard. Shipment delivery is also geofenced to help ensure drivers deliver to the correct location. Drivers attempting to deliver outside a geofence must correct their location or contact Dispatch to resolve the discrepancy.

Jitsu maintains a curated geocode and geofence database powered by a multi-provider geocoding system with nine geocoding providers arranged in a fallback chain. The system classifies addresses by type (residential, military, school, multi-dwelling unit, etc.) and caches validated results for up to one year. This address intelligence draws from public sources, private providers, and Jitsu's own delivery history.

Proof of Delivery (POD)

Drivers must take photo proof of delivery for every shipment without exception. PODs must include a photo of the parcel at the expected delivery location (doorstep, mailroom, parcel locker, etc.) as well as visible address, building, or unit number numerals. Jitsu uses a YOLO11-based package detection model running on-device via ONNX Runtime to assess whether the POD contains the expected elements in real time, without network dependency. A parallel OCR pipeline (DBNet + CRNN) extracts text for address and unit number verification. Image quality is also assessed on-device so drivers get immediate feedback to retake unusable photos. POD validation rules are configurable per client using an ANTLR-based domain-specific language, allowing client-specific requirements without code changes. Cloud-based ML services (Google Cloud Vision, Gemini VLM) provide additional batch analysis for complex POD interpretation and external review workflows.

Route Monitoring and Dispatch

The Jitsu system monitors each driver's route progress throughout the day. If a driver appears to be running dangerously behind or has stalled for an undue amount of time, the system progressively escalates: push notifications escalate to SMS, which escalate to phone calls from the Dispatch team. Drivers are expected to complete their routes within a defined time frame after pickup. Drivers who fail to do so receive reduced ratings and fewer future booking opportunities.

The Jitsu Dispatch app shows route status and allows dispatchers to follow progress throughout the day. If a route is in trouble, dispatchers can "rescue" it by splitting off a portion and assigning it to another driver, who arrives at a designated rescue location to take over the undelivered shipments. This mechanism helps protect OTD.

Fraud Detection

Gig-economy delivery creates fraud opportunities that traditional carriers do not face. Jitsu uses a multi-layered detection system with five algorithms, each targeting a different fraud pattern: cost anomaly detection (flags routes where cost per package exceeds normal thresholds), impossible travel detection (compares actual vs. expected travel time between stops), completion time analysis (detects routes completed suspiciously fast), dwell time analysis (identifies "drive-by" completions where the driver spent abnormally short time at stops), and POD daylight validation (checks POD photo timestamps against local sunrise/sunset times).

In addition, six real-time risk detectors monitor operations as they happen: mis-sort detection (catches packages scanned onto wrong routes within seconds), GPS signal monitoring (flags potential location spoofing), unauthorized departure detection (alerts when drivers leave the warehouse before scan completion), and new-driver monitoring (flags first-time drivers for additional oversight). Detection thresholds are configurable per region and driver tier via a Drools rule engine, without code deployment. Flagged routes are automatically triaged - high-risk items generate Slack alerts for manual review, while lower-risk items are auto-resolved with metrics tracking.

Customer Experience

Tracking

Recipients are informed of their delivery status via a tracking page. The tracking page shows shipment milestones (from ingestion to delivery), current status, and a live-updating ETA (refreshed every five minutes based on driver location after pickup).

Manage Delivery

The tracking page includes "Manage Delivery," a feature that allows recipients to confirm or update their address, delivery instructions, geocode pin, access codes, and delivery photos. This information is shown to drivers when the recipient's delivery is the next stop. Historic delivery photos are particularly useful for disambiguating delivery locations that look similar.

Changes made to delivery details - by either the Jitsu Dispatch team or recipients - are reflected in real time in the driver app.

SMS Communication

Jitsu uses SMS to notify recipients of delivery milestones and to solicit confirmations or updates for delivery information that appears missing or ambiguous (for example, apartment numbers or access codes). This proactive outreach helps maximize first-attempt delivery success rate.

Jitsu also uses SMS to communicate critical information to drivers, such as booking notifications.

Driver-Recipient Communication

Drivers can text or call recipients through a secure anonymizing proxy to resolve delivery issues such as missing access codes, access confusion, or address confusion. This is more responsive than the "call tag" mechanism used by national carriers.

Platform Apps and Services

The Jitsu platform consists of a number of web apps, mobile apps, and API services. Many of these have been referenced earlier in this document in the context of the operations they support. This section provides a consolidated overview.

Web Apps

- **Dispatch**: the primary tool for monitoring the day's routes, reviewing historic completed routes, and planning future routes. It shows all shipments on a filterable map (by region, driver type, or other categories). Clicking a route displays its shipments (customer names, addresses, sequence positions, etc.). Clicking a shipment shows full details including PODs, event history, and other metadata.
- **Inbound**: displays the sort layout for a given warehouse on a given day based on the routing solution. Used to monitor sort progress and performance (pieces per hour, or PPH). Can also be used to manually create sort layouts when the automated system is not handling layout generation for a given region on a given day.
- **Small Sort**: similar to the Inbound app but focused on the bin architecture (small parcel sorting) for a given warehouse on a given day based on the routing solution.
- **Outbound Dashboard**: used by Outbound teams to track expected and arriving drivers at a delivery station - their arrival times, vehicle types, and what kind of route needs to be staged for them.
- **Routing**: used by transportation planners to create routing problems and compute solutions using Jitsu's proprietary clustering, allocation, traffic, and routing algorithms.
- **Admin Dashboard ("Dashboard")**: allows administrators to configure various aspects of Jitsu's systems, including regions, clients, SLAs, warehouses, booking parameters, DSPs, driver programs, and more.
- **DSP Portal**: used by DSPs to manage and track their allocated routes and delivery driver fleet.
- **Recipient Portal**: the customer-facing tracking page that displays shipment status and provides the Manage Delivery feature.
- **Payment**: used to review IC and DSP payments, manage abnormal routes (routes flagged due to excessive delivery failures or other suspicious conditions), and configure payment formulas.
- **Client Portal**: used by Jitsu's shippers (clients) to track the progress of their shipments, including complete delivery history.

Mobile Apps

- **Jitsu Drive** ("the driver app"): guides drivers through registration, background checks, setting route preferences (used in conjunction with driver ratings to determine what delivery opportunities are shown, and when), booking routes, picking up routes, delivering routes, and receiving payment.
- **Jitsu Outbound** ("the outbound app"): used by outbound staff to manage yard parking spots, track arriving drivers, and bring routes to drivers ("check out" routes). Complements the Outbound Dashboard web app.
- **Jitsu Warehouse**: used by sortation workers to receive inbound containers, pre-sort shipments (first scan), sort large parcels, sort small parcels, and audit that parcels have been sorted to the correct pallet position or bin.

API Services

- **Public API**: the service integrated with by shippers and shipping platforms (such as Shippo or EasyPost) for rate quoting, submitting shipments, generating labels, tracking shipments, canceling shipments, and exchanging or updating shipment information generally.
- **Driver API**: the backend service that supports the Jitsu Drive mobile app.

Platform Technology

This section describes the technology underpinnings of the Jitsu platform.

Technology Stack

Backend services are primarily written in Java (Dropwizard and Quarkus frameworks), with Python (Flask, Celery) for ML and data services and Kotlin (Ktor) for some newer services. Internal service-to-service communication uses gRPC. Business rules (booking eligibility, pricing, shipment actions) are implemented via the Drools rule engine. The pricing DSL and POD validation DSL use ANTLR for parser generation.

Mobile apps are built in Flutter/Dart, enabling cross-platform deployment. Web applications are built in React.js (with TypeScript in newer apps), with some specialized apps using Vue.js.

The data layer uses PostgreSQL (primary relational store), MongoDB (document store), Apache Cassandra (event persistence and time-series data), Redis (caching, distributed locks, driver rating rankings), Elasticsearch (search), and Google BigQuery (analytics and reporting). Object storage uses S3 and Google Cloud Storage.

ML and AI capabilities include YOLO v11 and ONNX Runtime for on-device inference, Google Cloud Vision and Gemini VLM for cloud-based image analysis, XGBoost and SARIMA for demand forecasting, and Vertex AI RAG for operational knowledge retrieval.

The routing and mapping stack includes Google OR-Tools, OptaPlanner, and a forked version of jsprit (open-source VRP library extended with Jitsu-specific customizations), GraphHopper for traffic-aware routing, and Uber H3 hexagonal grid indexing for geospatial operations.

Infrastructure runs on Google Cloud Platform, with Consul for service discovery and Datadog for observability.

Event-Driven Architecture

Every meaningful action in the Jitsu platform emits an event. The system processes over 1 billion events per year across 392 defined event types, handled by 213 specialized worker processes. Events are routed via RabbitMQ and persisted to Cassandra.

High-volume event categories include webhooks to shippers (265M/year), delivery status updates (218M), package scans (177M), driver location updates (152M), and POD photos (90M).

A single operational event (such as "package delivered") can fan out to five or more downstream workflows: customer notification, driver rating update, webhook to the shipper, fraud check, and billing calculation. Workers follow standardized patterns for scheduling, distributed locking, and error handling, with built-in batching and throttling to prevent downstream overload during peak periods.

This architecture provides a complete audit trail - every event includes timestamp, actor, payload, and causation chain - enabling debugging, regulatory compliance, and dispute resolution without forensic investigation.

Offline-First Driver App

The Jitsu Drive mobile app is engineered to operate reliably in varying network conditions, a critical capability for last-mile delivery where drivers frequently encounter dead zones, underground parking garages, and building interiors.

All delivery events are queued locally in Hive storage with automatic upload when connectivity returns. Complete route data is cached locally so drivers can continue deliveries during network outages with full stop details, navigation, and package information. POD photos are stored locally with background sync to cloud storage, handling interruptions, duplicates, and upload failures gracefully. App state is persisted so the app recovers seamlessly from crashes, backgrounding, and restarts without losing active delivery progress.

The app also includes anti-tampering measures: root/jailbreak detection, fake location detection, and NTP-based time validation (via a custom true-time library) to prevent clock manipulation of event timestamps.